Dhirubhai Ambani University
Technology

Formerly DA-IICT

Advanced Statistical Tools for Data Science

Assignment-3 : Simple Regression with Observational Data, Missing Data, and Causal Reasoning

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Walking Activity and Body Weight

GitHub Repository:
`https://github.com/atik-7866/Advanced-Stat`

Group Number: 6

Aaditya Pala (202301076)
Atik Vohra (202301447)
Aum Patel (202301448)
Vivek Parmar (202301475)

Introduction

This project analyses the relationship between daily step count (recorded via smartphones or fitness wearables) and body weight (kg). Data were collected from students using a Google Form. We applied simple linear regression to examine whether step count affects body weight, using a scatter plot to visualise the data, a QQ plot to assess normality, and Cook's distance to detect influential observations. For missing data analysis and causal reasoning via a Directed Acyclic Graph (DAG), we took assistance from a Large Language Model (LLM).

Question: Is there an association between daily walking steps and body weight (kg)?

- **Predictor** (x): Average steps per day in thousands (recorded by phone or fitness tracker)
- **Response** (y): Body weight (kg)

Data Collection

Method of Data Collection

We have collected data from students through a Google Form and asked them to fill in their age, gender, the average number of steps they walk per day (as shown on their phone or smart watch), their body weight, their daily step goal, and their daily work and exercise routine. This data is an important part of our analysis and causation; based on this, our results are affected.

Plot

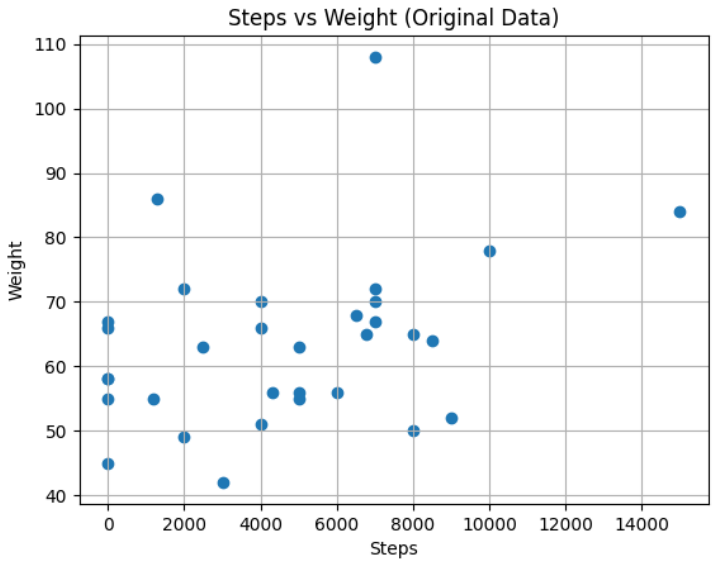

Figure 1: Scatter Plot: Steps per Day vs Body Weight (kg). Some students reported no entries for steps, shown on the y-axis.

Descriptive Statistics of Data

Table 1 shows the summary statistics for the full dataset ($n = 32$).

Table 1: Summary Statistics (Full Dataset, $n = 32$)

Statistic	Steps per Day	Weight (kg)
Count	32	32
Mean	4658.81	63.50
Std Dev	3591.08	13.12
Min	0	42.00
25%	1824.75	55.00
50%	4656.00	63.50
75%	7000.00	68.50
Max	15000	108.00

The average daily step count is about 4659 steps and the average body weight is 63.5 kg. There is a large spread in step counts ($\sigma = 3591$), indicating diverse activity levels among participants.

Regression Model

We fit a simple linear regression model of the form:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where y is body weight (kg) and x is steps per day **in thousands** (i.e. $x = \text{steps}/1000$).

Fitted Model (After Removing Missing Values, $n = 26$)

Using all 26 complete observations, the estimated model is:

$$\hat{y} = 57.03 + 1.343\, x$$

Table 2: Regression Model Summary ($n = 26$)

Quantity	Value
Intercept $\hat{\beta}_0$	57.031
Slope $\hat{\beta}_1$	1.343
R^2	0.0898
Adjusted R^2	0.0519
Correlation Coefficient r	0.2997
Standard Error of Estimate S_e	13.502
Standard Error of $\hat{\beta}_0$, S_{b_0}	5.660
Standard Error of $\hat{\beta}_1$, S_{b_1}	0.872
p-value (slope)	0.137
95% CI for $\hat{\beta}_0$	(45.94, 68.12)
95% CI for $\hat{\beta}_1$	(−0.367, 3.053)

Remarks:

- The intercept $\hat{\beta}_0 = 57.03$ means that when steps $= 0$, the estimated body weight is about 57 kg.
- The slope $\hat{\beta}_1 = 1.343$ means that for each additional 1000 steps per day, body weight increases by approximately 1.343 kg on average. However, this effect is not statistically reliable.
- $R^2 \approx 0.090$: only about 9% of the variation in body weight is explained by step count.
- p-value $= 0.137$: since $0.1 < p \leq 0.2$, the evidence against the null hypothesis is weak. The slope is statistically insignificant.
- The correlation $r = 0.30$ indicates a weak positive linear relationship.
- The 95% CI for $\hat{\beta}_1$ is $(-0.367,\ 3.053)$, which includes zero, further confirming the slope is not significant.

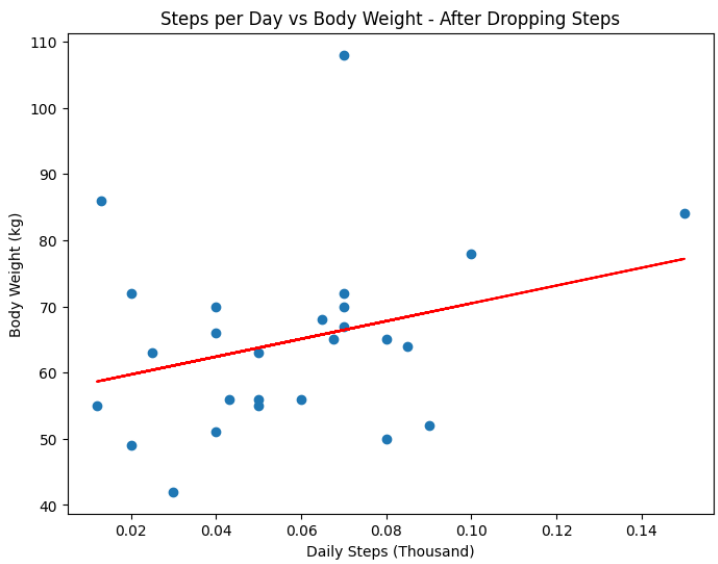

Figure 2: Steps per Day vs Body Weight with Fitted Regression Line

Residual Analysis

Standardised Residuals Plot

The standardised residual for observation i is:

$$e_i^* = \frac{e_i}{S_e}, \qquad \text{where } e_i = y_i - \hat{y}_i$$

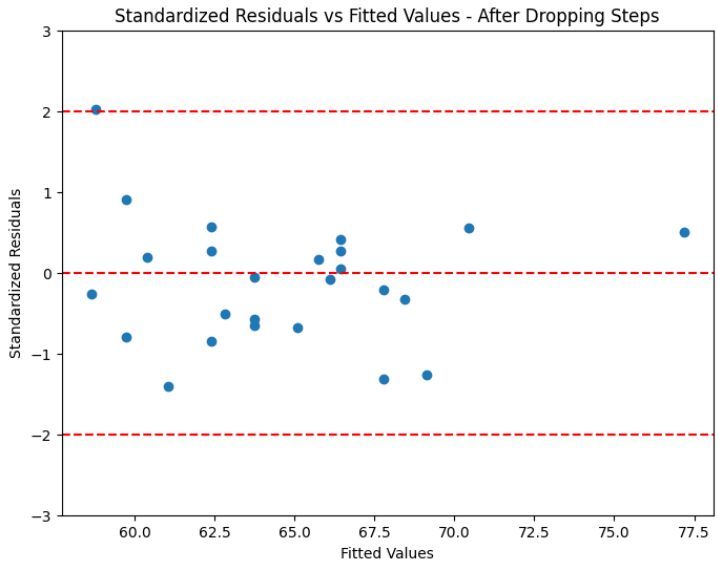

Figure 3: Standardised Residuals vs Fitted Values

Remarks:

- Most points are close to zero.
- Points are spread on both sides of zero, so no strong pattern is visible.
- One point is slightly above +2, indicating a mild outlier.

QQ Plot (Normality Check)

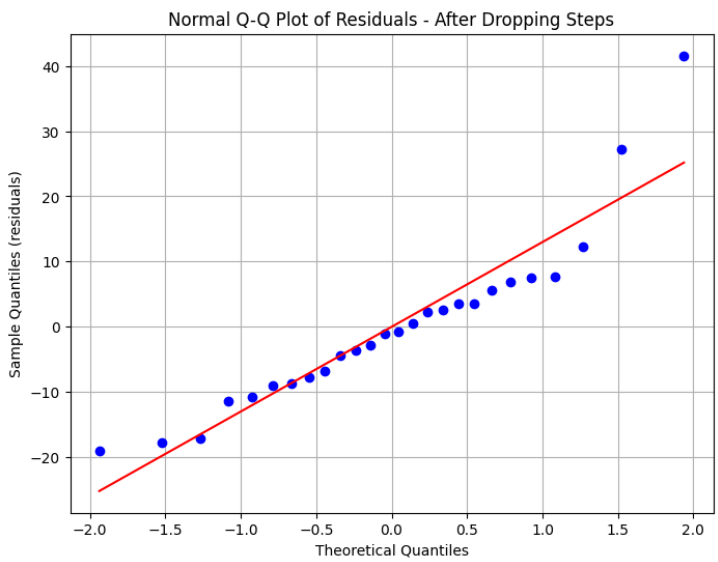

Figure 4: Normal QQ Plot of Residuals

Remarks:

- Most points in the middle follow the straight diagonal line.

- Points on the right side and left side deviate above the line, indicating a heavier right tail.
- This shows the residuals are not perfectly normal.
- A few extreme values are affecting the distribution.

Cook's Distance

Cook's distance C_i measures how much the fitted values change when observation i is removed:

$$C_i = \frac{\sum_{j=1}^{n} \left(\hat{y}_j - \hat{y}_{j(i)}\right)^2}{S_e^2 \, (p+1)}$$

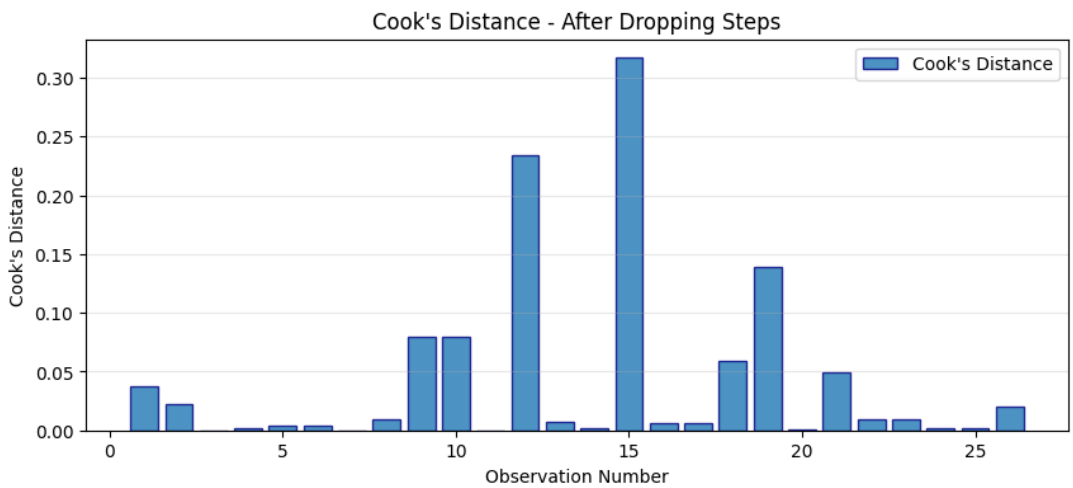

Figure 5: Cook's Distance for Each Observation

Table 3: Most Influential Observations by Cook's Distance

Obs	**Steps**	**Weight (kg)**	**Cook's C_i**
15	1299	86	0.317
12	7000	108	0.235
19	15000	84	0.140

Remarks:

- Observation 15 (1299 steps, 86 kg) has the highest Cook's distance ($C_{15} = 0.317$). This person has very few steps but high weight – the opposite of the general trend.
- Observation 12 (7000 steps, 108 kg) is the second most influential ($C_{12} = 0.235$). This person has average steps but very high weight.

Effect of Removing Influential Observations

After Removing One Influential Point (Obs 15)

Removing observation 15, the new model becomes:

$$\hat{y} = 52.55 + 1.916\,x$$

Table 4: Model After Removing Obs 15 ($n = 25$)

Quantity	**Original** ($n = 26$)	**After Removing Obs 15** ($n = 25$)
Intercept $\hat{\beta}_0$	57.031	52.553
Slope $\hat{\beta}_1$	1.343	1.916
R^2	0.0898	0.1854
p-value (slope)	0.137	0.032
95% CI for $\hat{\beta}_1$	$(-0.367,\ 3.053)$	$(0.275,\ 3.557)$

Remarks:

- R^2 increased from 9.0% to 18.5% – the model now explains nearly twice the variation.
- The slope increased from 1.343 to 1.916, showing a stronger positive effect of steps on weight.
- p-value $= 0.032$: since $0.01 < p \leq 0.05$, the evidence against the null hypothesis is moderately strong. The slope is now statistically significant at the 5% level.
- The 95% CI for $\hat{\beta}_1$ is $(0.275,\ 3.557)$, which no longer includes zero.

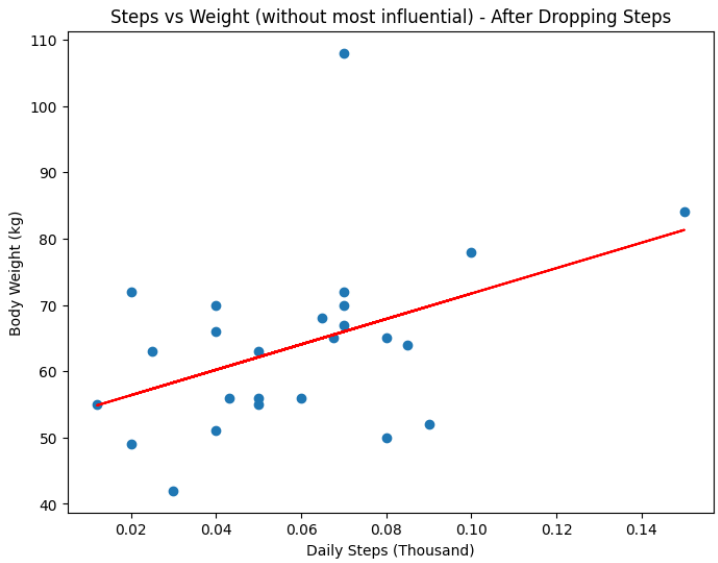

Figure 6: Model after Removing One Influential Point (Obs 15)

After Removing Two Influential Points (Obs 15 and 12)

Removing both observations 15 and 12, the model becomes:

$$\hat{y} = 52.09 + 1.697\,x$$

Table 5: Model After Removing Obs 15 & 12 ($n = 24$)

Quantity	**Original** ($n = 26$)	**Remove Obs 15** ($n = 25$)	**Remove Obs 15 & 12** ($n = 24$)
Intercept $\hat{\beta}_0$	57.031	52.553	52.085
Slope $\hat{\beta}_1$	1.343	1.916	1.697
R^2	0.0898	0.1854	0.2717
p-value (slope)	0.137	0.032	0.009
95% CI for $\hat{\beta}_1$	$(-0.367, 3.053)$	$(0.275, 3.557)$	$(0.536, 2.858)$

Remarks:

- R^2 improved further from 18.5% to 27.2% – a significant gain in model fit.
- p-value $= 0.009$: since $0.001 < p \leq 0.01$, the evidence against the null hypothesis is strong. The result is clearly significant.
- The slope is slightly lower (1.697) than after removing only one point, but the model is more stable.
- Overall, removing the two influential outliers reveals a much clearer positive relationship between steps and body weight.

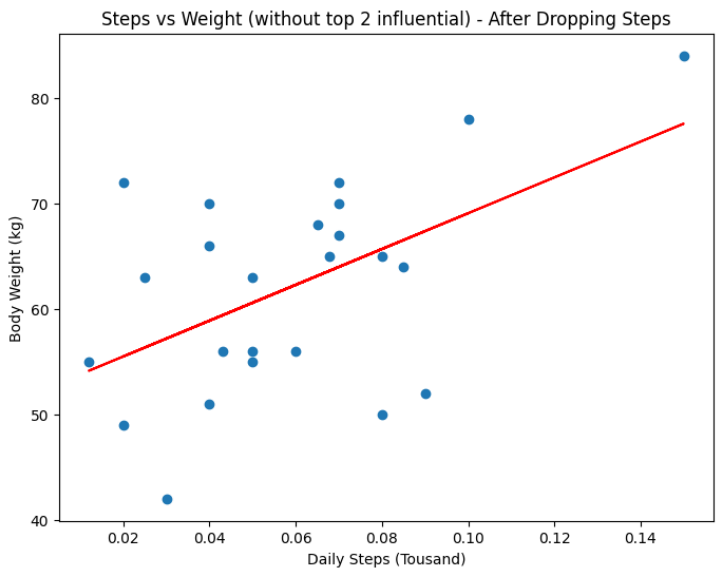

Figure 7: Model after Removing Two Influential Points (Obs 15 & 12)

Missing Data Analysis

Method Used: Routine Based Random Imputation

Since the missing step values are related to each student's self-reported daily routine, we use a **routine-based random imputation** strategy. The logic is:

- If a student reported a *physically active* routine, their missing steps are filled with a random value drawn uniformly from [10000, 15000].
- If a student reported a *mostly sitting at desk* routine, missing steps are filled from [2000, 5000].
- For any other routine, missing steps are filled from [1000, 15000].

This is preferable to plain mean imputation because it uses available information (the `Routine` variable) to generate more realistic and varied step values, rather than collapsing all missing entries to a single number.

After Routine Based Random Imputation

Fitted Model ($n = 32$)

After filling missing step values using routine-based random imputation, the estimated model is:

$$\hat{y} = 57.63 + 1.007\,x$$

Table 6: Regression Model Summary : Routine Based Imputation ($n = 32$)

Quantity	**Value**
Intercept $\hat{\beta}_0$	57.628
Slope $\hat{\beta}_1$	1.007
R^2	0.0579
Adjusted R^2	0.0265
Correlation Coefficient r	0.2407
Standard Error of Estimate S_e	12.950
p-value (slope)	0.185
95% CI for $\hat{\beta}_1$	$(-0.446,\ 2.460)$

Remarks:

- The slope $\hat{\beta}_1 = 1.007$ is smaller than in 1.343, because the imputed values add noise and reduce the apparent effect.
- $R^2 = 0.058$: only 5.8% of variation explained – weaker than the dropped model.
- p-value $= 0.185$: since $0.1 < p \leq 0.2$, the evidence against the null hypothesis is weak. We do not reject H_0.
- The 95% CI for $\hat{\beta}_1$ is $(-0.446,\ 2.460)$, which includes zero.

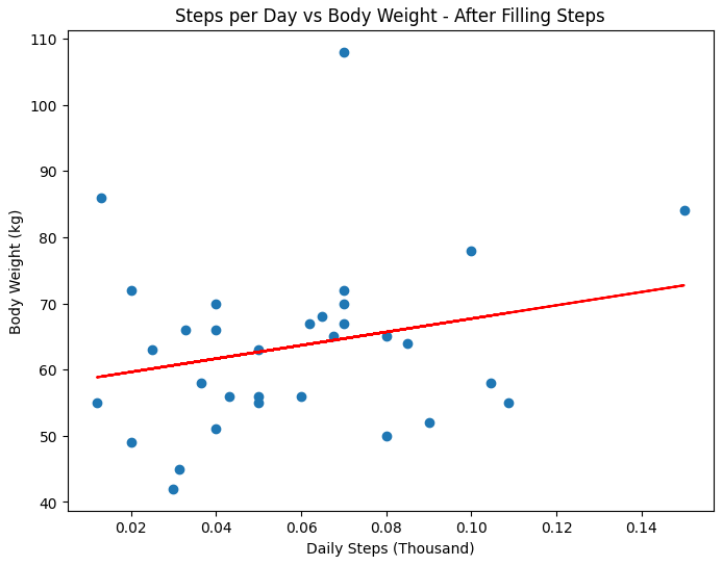

Figure 8: Fitted Regression Line after Routine Based Imputation

Standardised Residuals

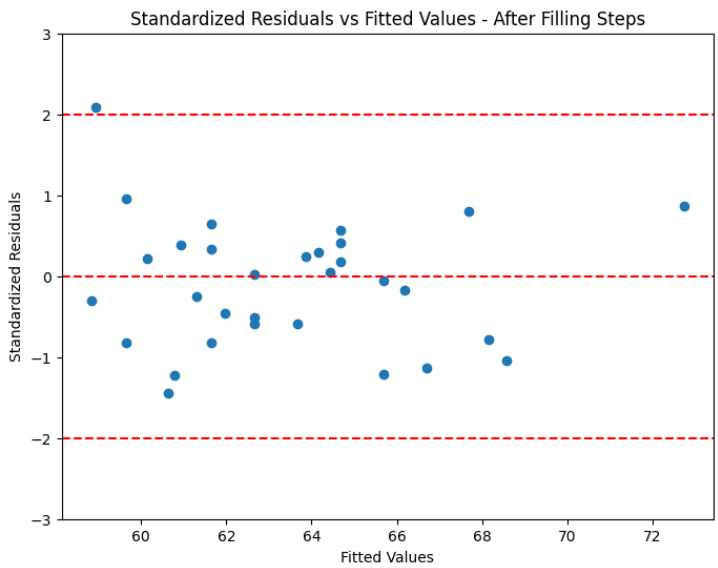

Figure 9: Standardised Residuals vs Fitted Values after Routine Based Imputation

Remarks:

- Most residuals lie within ± 2.
- The spread is similar to Dropping Missing Values, but imputed points are more distributed since routine-based values cover a range.
- No strong pattern, so constant variance is roughly maintained.

QQ Plot

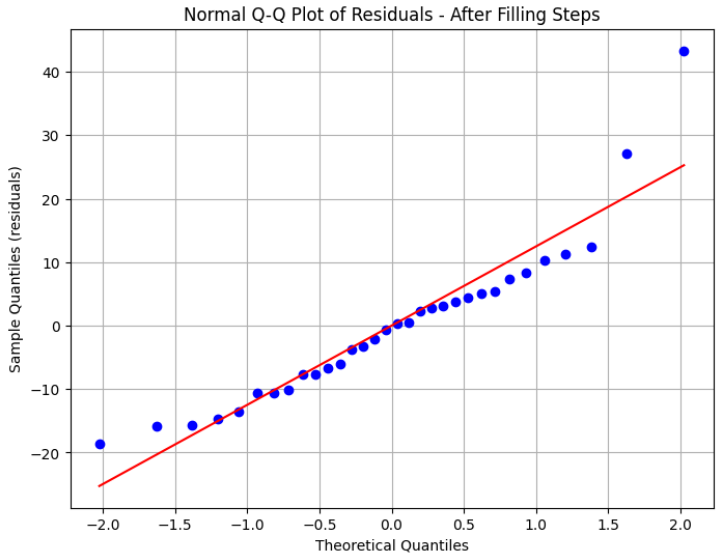

Figure 10: Normal QQ Plot of Residuals after Routine Based Imputation

Remarks:

- The middle points follow the diagonal line acceptably.
- The right tail still deviates, indicating a few heavy-tailed observations.

Cook's Distance

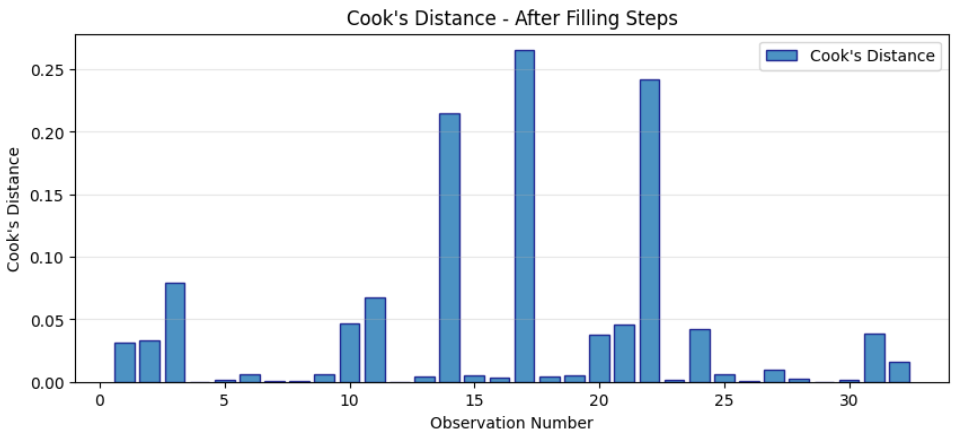

Figure 11: Cook's Distance for Each Observation after Routine Based Imputation

Table 7: Most Influential Observations by Cook's Distance

Obs	**Steps**	**Weight (kg)**	**Cook's C_i**
17	1299	86	0.265
22	15000	84	0.242
14	7000	108	0.215

Remarks:

- The same extreme individuals remain influential even after imputation.
- Obs 17 (1299 steps, 86 kg) is still the most influential ($C_{17} = 0.265$).
- Cook's distances are slightly lower than in Dropping Missing Values because the larger sample ($n = 32$) dilutes their influence.

After Removing One Influential Point (Obs 17, $n = 31$)

$$\hat{y} = 54.09 + 1.453\,x$$

Table 8: Effect of Removing Obs 17, after Routine Based Imputation

Quantity	**Full** ($n = 32$)	**Remove Obs 17** ($n = 31$)
Slope $\hat{\beta}_1$	1.007	1.453
R^2	0.0579	0.1244
p-value	0.185	0.052

Remarks:

- R^2 improved from 5.8% to 12.4%.
- p-value $= 0.052$: since $0.05 < p \leq 0.1$, the evidence against the null hypothesis is on the margin of strong and weak. The result is borderline significant.

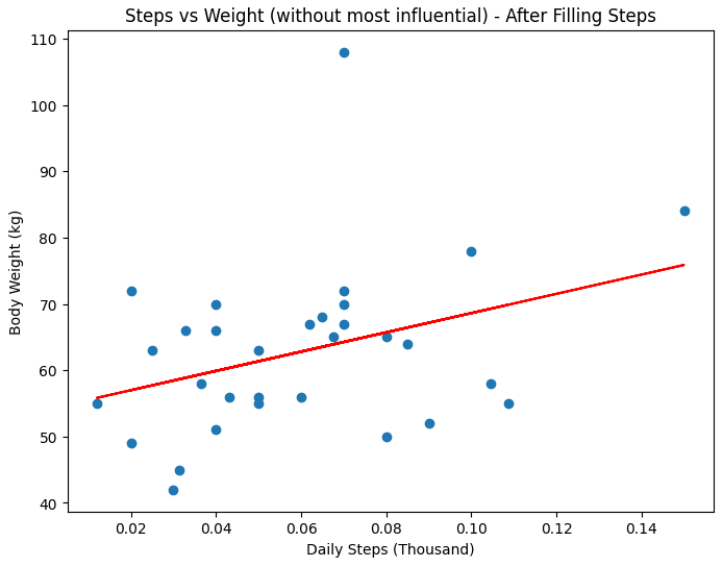

Figure 12: Regression Line after Removing Obs 17 , after Routine Based Imputation

After Removing Two Influential Points (Obs 17 & 22, $n = 30$)

$$\hat{y} = 55.97 + 1.074\,x$$

Table 9: Effect of Removing Obs 17 & 22 , after Routine Based Imputation

Quantity	**Full** ($n = 32$)	**Remove Obs 17** ($n = 31$)	**Remove Obs 17 & 22** ($n = 30$)
Slope $\hat{\beta}_1$	1.007	1.453	1.074
R^2	0.0579	0.1244	0.0530
p-value	0.185	0.052	0.221

Remarks:

- Removing the second influential point (Obs 22: 15000 steps, 84 kg) causes R^2 to drop back to 5.3% and p-value to increase to 0.221.
- p-value = 0.221: since $p > 0.2$, the evidence against the null hypothesis is moderately weak.
- This suggests Obs 22 was actually supporting the regression trend (high steps); removing it weakens the fit. It may not be a genuine outlier.

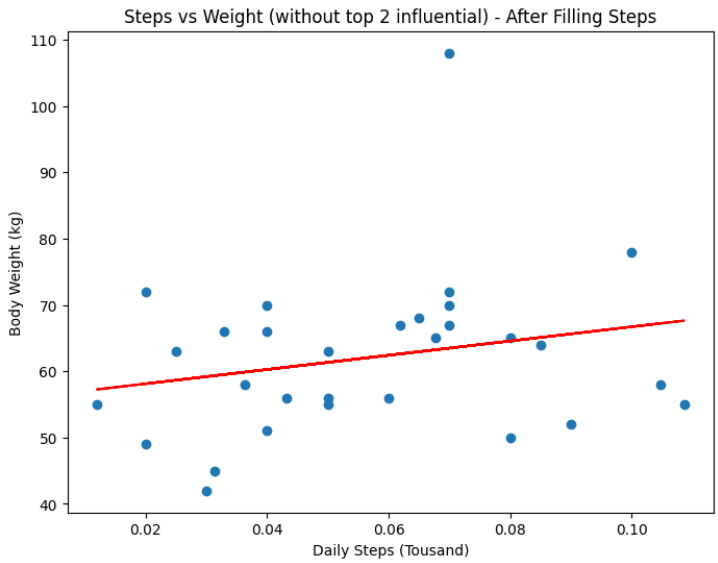

Figure 13: Regression Line after Removing Obs 17 & 22, after Routine Based Imputation

Comparison: Dropping Missing Value vs Routine-Based Imputation

Table 10: Summary Comparison: Dropping vs Routine-Based Imputation

Method	n	R^2	**Slope** $\hat{\beta}_1$	p**-value**	**Evidence**
Drop missing	26	0.090	1.343	0.137	Weak
Routine imputation	32	0.058	1.007	0.185	Weak
Remove 2 outliers(Drop missing)	24	0.272	1.697	0.009	Strong
Remove 1 outlier(Routine imputation)	31	0.124	1.453	0.052	Marginal

Key Takeaways:

- Dropping missing values gives a slightly stronger model than Routine Based Imputation because it avoids introducing noise.
- Routine-based imputation is better than plain mean imputation, as it preserves variation across activity groups.
- In both options, the model only becomes clearly significant after removing influential outliers.

Potential Bias Introduced

- **Dropping (listwise deletion):** If missingness is MNAR (sedentary people missing), the remaining sample over-represents active individuals, biasing the slope estimate upward.
- **Routine-based imputation:** If routine categories do not accurately reflect actual step counts, the imputed values introduce noise, reducing R^2 and attenuating the slope.

Causal Directed Acyclic Graph (DAG)

DAG Diagram

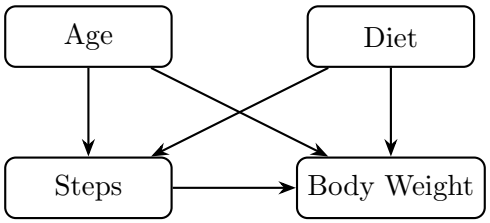

Figure 14: Causal DAG: Steps, Body Weight, Diet, and Age

Explanation of the DAG

- **Steps → Body Weight:** The direct causal path we are interested in. Higher daily steps may cause lower body weight through caloric expenditure.
- **Age → Steps:** Older individuals tend to walk fewer steps due to reduced mobility.
- **Age → Body Weight:** Age affects body weight through changes in metabolism, muscle mass, and hormonal factors.
- **Diet → Body Weight:** Diet is one of the strongest direct causes of body weight.
- **Diet → Steps:** Health-conscious people tend to both eat well and walk more.

Confounding

Age is a confounder: it affects both step count and body weight. If we do not control for age, the observed association may partly reflect the effect of age.

Diet is also a confounder: individuals with healthy diets tend to walk more and have lower weight. Not accounting for diet means the regression slope for steps absorbs some of diet's effect.

In our simple linear regression, both age and diet act as omitted confounders, likely biasing the estimated relationship.

Conclusion

- The fitted model $\hat{y} = 57.03 + 1.343\,x$ (where x is steps in thousands) shows a weak positive relationship. The slope is not statistically significant ($p = 0.137$, evidence is weak).
- $R^2 = 0.090$ indicates that step count explains only about 9% of the variation in body weight. The model has poor predictive power.
- Two influential observations (Obs 15: 1299 steps, 86 kg and Obs 12: 7000 steps, 108 kg) distort the regression. After removing them, R^2 improves to 27.2% and the evidence becomes strong ($p = 0.009$).
- Missing data is most likely MAR (related to observed routine). Routine-based imputation is used; however it slightly weakens the model compared to listwise deletion due to added noise.
- The DAG reveals that diet and age are confounders. Simple regression of weight on steps alone cannot establish a causal effect without controlling for these variables.
- The sample size is small, so the data is not enough to build a strong model.